

SILAS KWABLA GAH

PhD Candidate in Computer Science · Deep Learning, World Models & Test-Time Adaptation

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Research Profile

Final-year PhD candidate in Computer Vision and Machine Learning at the University of Ghana, working on geometric 3D scene understanding, multimodal foundation models, test-time adaptation ($\Delta\theta = 0$), and reinforcement learning for grounded visual reasoning. Research focuses on inference-time coordination of independently operating visual models, camera-geometry-based 2D–3D lifting, semantic evidence preservation, and cross-view consistency as self-supervised and process-level oversight signals. Experienced with large-scale 3D/video point-cloud evaluation, PyTorch GPU acceleration, and memory-bounded inference pipelines.

Education

PhD in Computer Vision and Machine Learning

University of Ghana, Legon

Expected defence: October 2026

Supervisor: Prof. Ebenezer Owusu

Thesis: *Visual Perception under Low Supervision: Inference-Time Coordination of Frozen Foundation Model Representations*

Research investigates frozen foundation-model composition, open-vocabulary 2D/3D segmentation, semantic evidence preservation (avoiding premature collapse), multi-view geometric consensus, and reliability-aware inference.

MPhil in Computer Science

University of Ghana, Legon

2016

Accra, Ghana

BSc in Computer Science

Ghana Institute of Management and Public Administration (GIMPA)

2012

Accra, Ghana

HND in Electrical & Electronic Engineering

Accra Technical University

2009

Accra, Ghana

Research Experience

PhD Research — 3D Scene Understanding, Foundation Models & Test-Time Coordination University of Ghana

- Formulated **Reliability-Calibrated Adaptive Semantic Arbitration (RCASA)**, an inference-time framework that coordinates heterogeneous foundation models without gradient updates ($\Delta\theta = 0$), combining Platt calibration, uncertainty modeling, and contextual reliability weighting.
- Identified **premature semantic collapse** as a fundamental information bottleneck in multi-model composition; designed controlled Top- k retention experiments demonstrating that preserving secondary semantic hypotheses substantially improves multi-source coordination.
- Developed a camera-geometry-based **2D-to-3D lifting and consensus pipeline** using calibrated intrinsics, extrinsic poses, and multi-view semantic accumulation to reconcile dense 3D vision-language predictions (**RegionPLC**) with projected 2D promptable segmentations (**SAM3**).
- Evaluated across benchmark protocols on **ScanNet200** (312 scenes, $\sim 50M$ points), **ScanNet++**, and alternative sparse sources (**GroundingDINO+SAM2.1**) using paired scene-level bootstrap confidence intervals.
- Built streaming and memory-bounded infrastructure for evaluating dense per-point semantic distributions via chunked processing, float16 caching, sparse tensors, and calibration diagnostics.

Multimodal RL — Cross-View Agreement as a Process-Level Reward

University of Ghana

- Developed a process-level grounding reward leveraging consensus between frozen 3D and image-based perception sources, converting cross-view physical consistency into self-supervision without human-annotated reward labels.
- Built a **GRPO/VERL** multimodal RL pipeline for **Qwen2.5-VL-7B-Instruct** with **vLLM** rollout generation and structured grounding rewards; completed reward-validation and counterfactual discrimination studies against specification gaming.
- Evaluated reward dynamics across 1,000+ multimodal reasoning rollouts, separating spatial grounding validity from final-answer correctness and eliminating area-bias gaming ($\rho > 0.72 \rightarrow 0.334$).

Selected Publications and Manuscripts

- Training-Free Open-Vocabulary 3D Point-Cloud Segmentation on the Generalized Few-Shot Benchmark.**
ACCV 2026, Paper #482; preprint available. First Author.

- **Training-Free Generalised Few-Shot Segmentation through Open-Vocabulary Semantic Arbitration.**
WACV 2027, Paper #167. First Author.
- **Beyond Argmax: A Mechanistic Study of Semantic Retention in Frozen Foundation-Model Composition for Generalized Few-Shot 3D Segmentation.**
Manuscript in preparation (targeted for TPAMI/ICLR). First Author.
- **Cross-View Agreement as a Label-Free Process Reward for Multimodal Reasoning.**
SACAIR 2026 submission. First Author.

Technical Skills

Programming	Python, Java, C++
Deep Learning / RL	PyTorch, Transformers, vLLM, LoRA/QLoRA, SFT, GRPO, VERL
GPU & Systems	CUDA fundamentals, Triton kernel development and benchmarking, GPU profiling, memory-efficient inference and evaluation pipelines
3D & Geometric Vision	Point clouds, camera intrinsics/extrinsics, coordinate transforms, posed-view projection, 2D–3D lifting, multi-view semantic aggregation
Foundation Models	RegionPLC, SAM3, SAM2.1, GroundingDINO, CLIP, DINOv2, Qwen2.5-VL
Methodology	Paired bootstrap confidence intervals, calibration analysis (ECE, Brier), controlled ablations, matched causal interventions, large-scale evaluation

Teaching, Professional Experience & Service

Part-time Lecturer	Accra Technical University (2022–Present)
• Lecturing in computing, algorithms, and applied machine learning; previously taught at GIMPA.	
Founder & Lead Architect	EED Soft Consult (2018–Present)
• Architected and delivered large-scale, production-grade distributed and mobile software systems for enterprise and public-sector clients in Ghana.	
Academic Service	Reviewer, SACAIR 2026 Program Committee

Research Interests

World models; collaborative perception; test-time adaptation; self-supervised learning; 3D scene understanding; foundation-model composition; embodied AI; multimodal reinforcement learning.